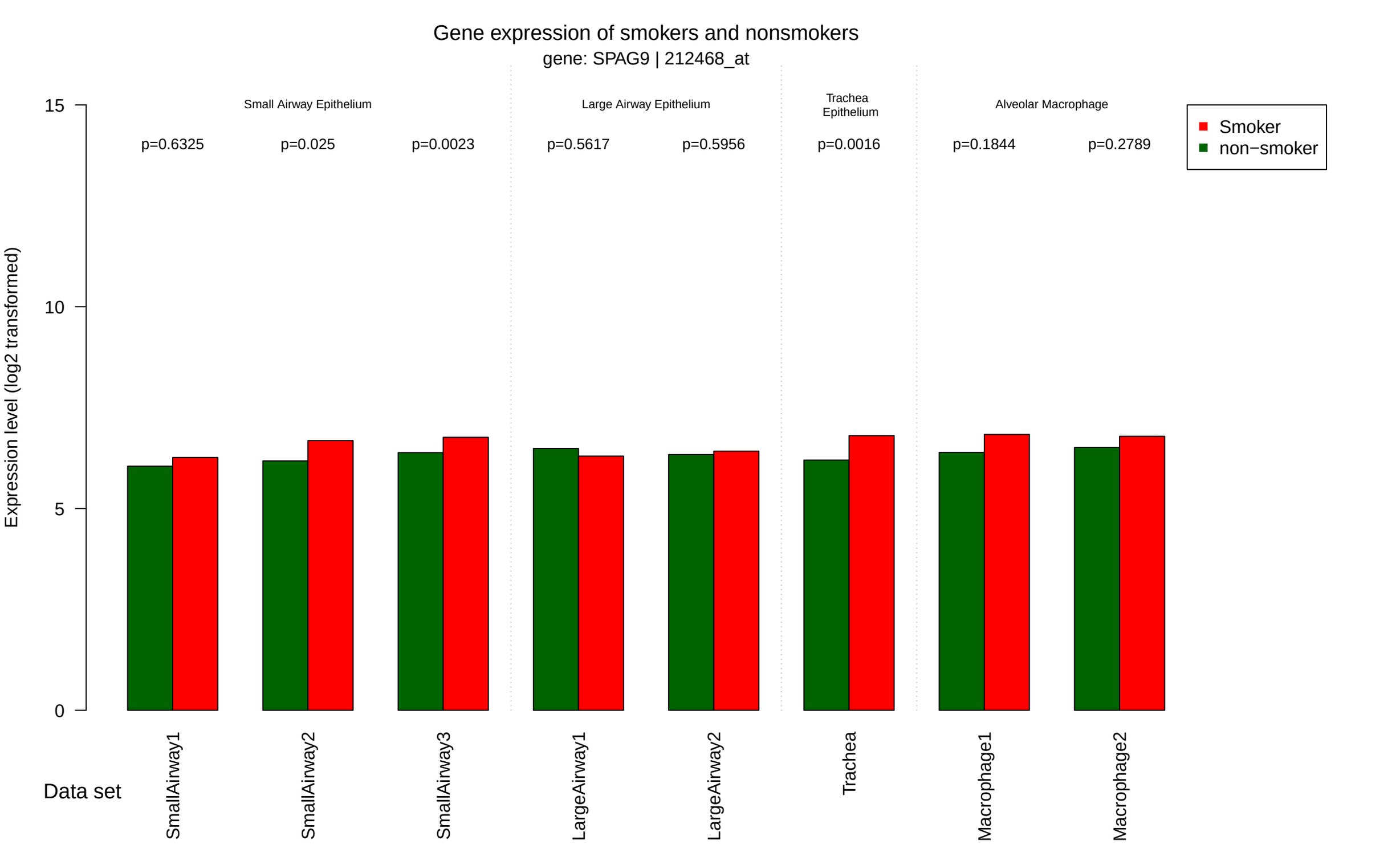
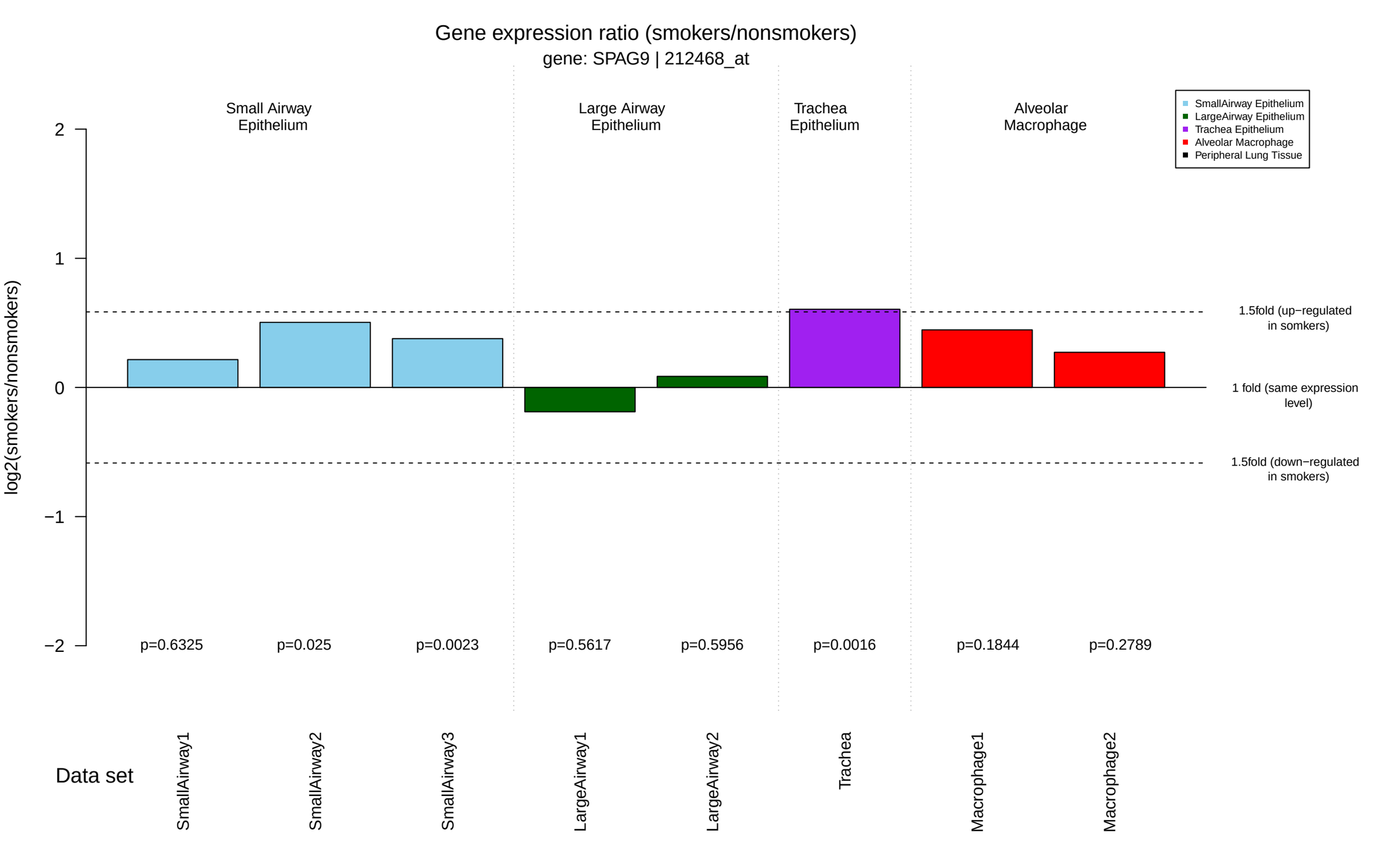
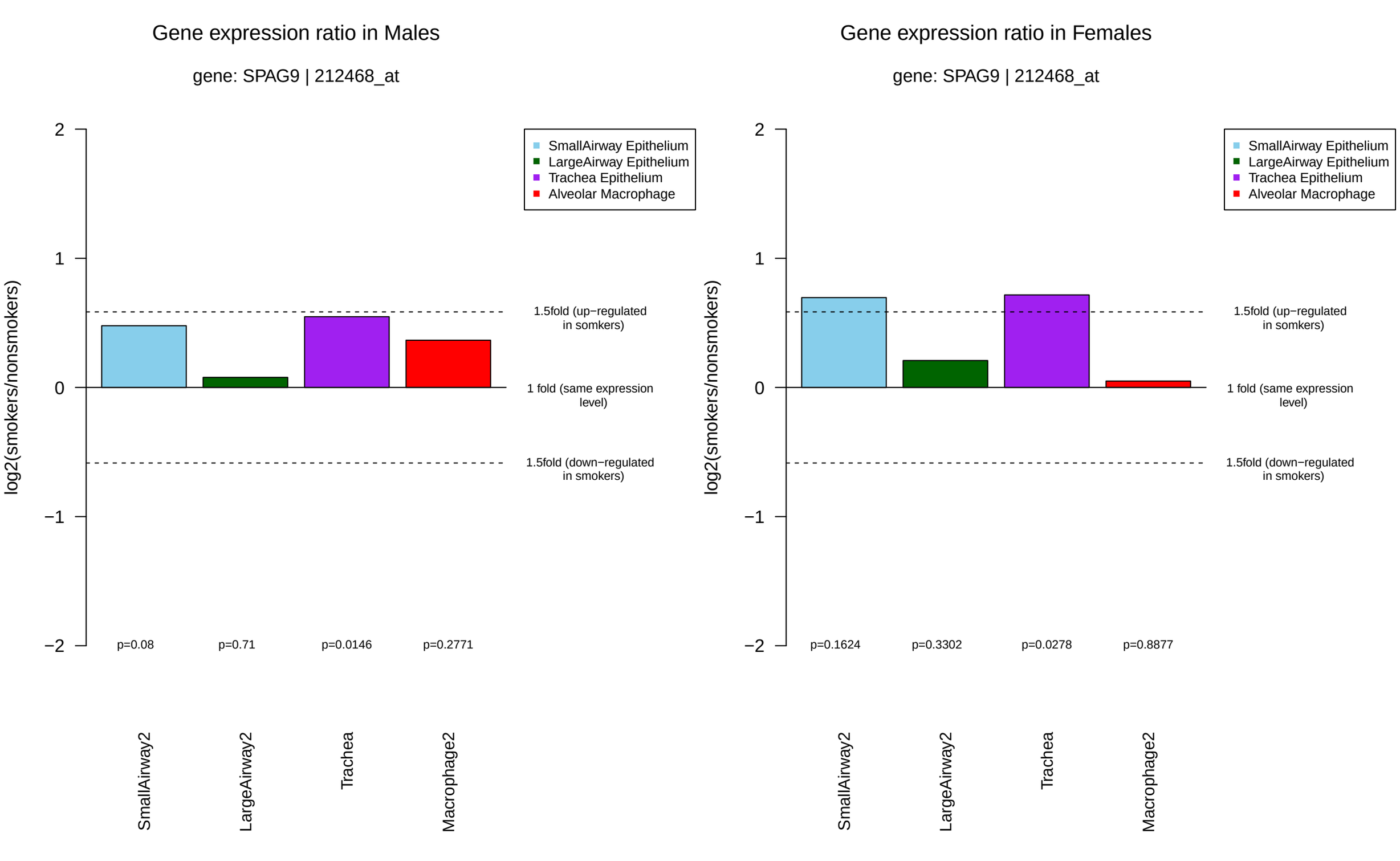
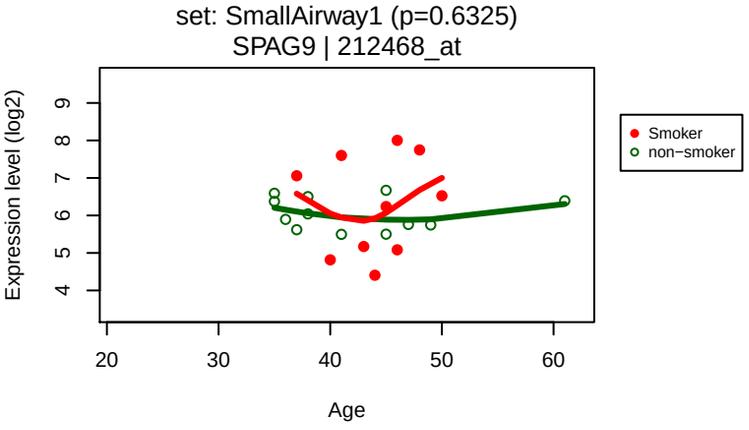
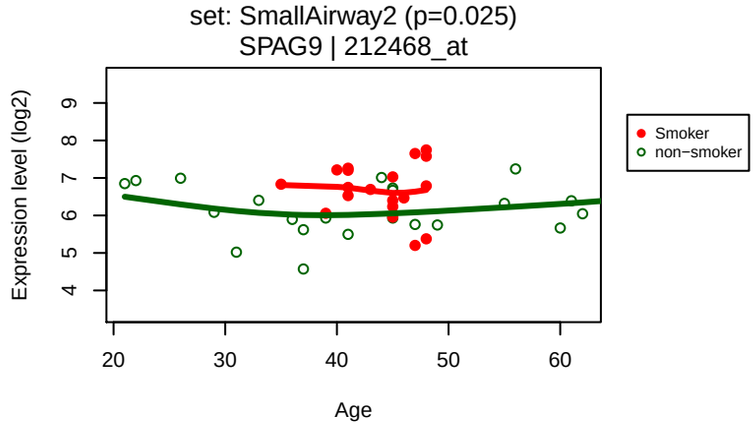
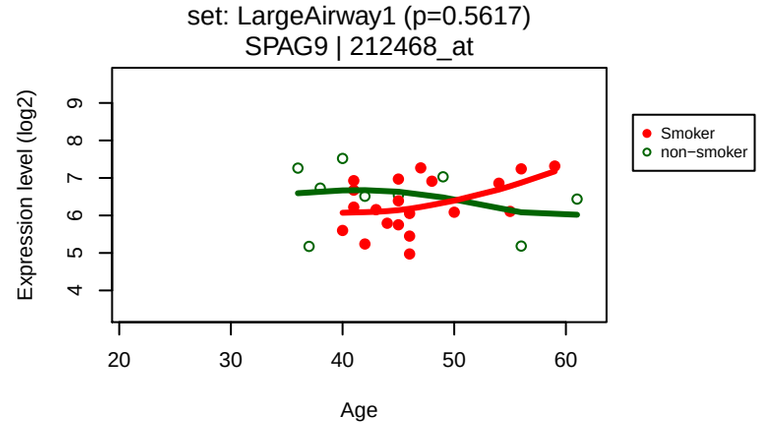

Gene expression of smokers and nonsmokers
gene: SPAG9 | 212468_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6325
p=0.025
p=0.0023
p=0.5617
p=0.5956
p=0.0016
p=0.1844
p=0.2789
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

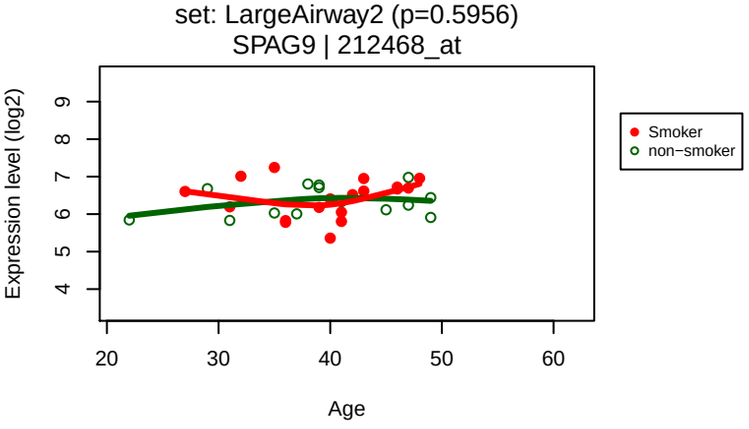

Gene expression ratio (smokers/nonsmokers)
gene: SPAG9 | 212468_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6325
p=0.025
p=0.0023
p=0.5617
p=0.5956
p=0.0016
p=0.1844
p=0.2789
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

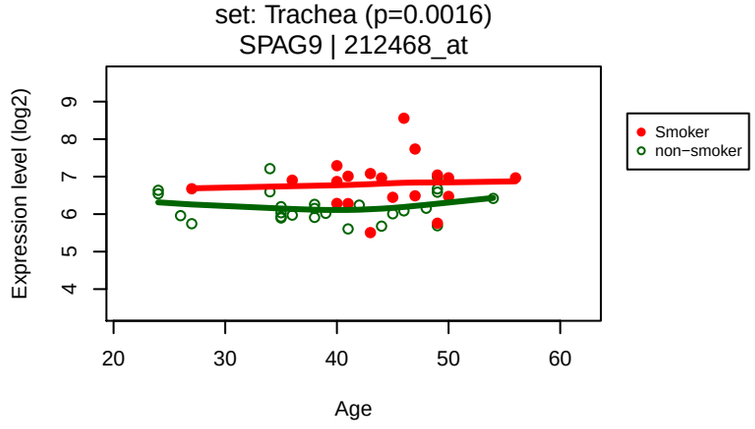

Gene expression ratio in Males
gene: SPAG9 | 212468_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.08
p=0.71
p=0.0146
p=0.2771
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPAG9 | 212468_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1624
p=0.3302
p=0.0278
p=0.8877
SmallAirway2
LargeAirway2
Trachea
Macrophage2

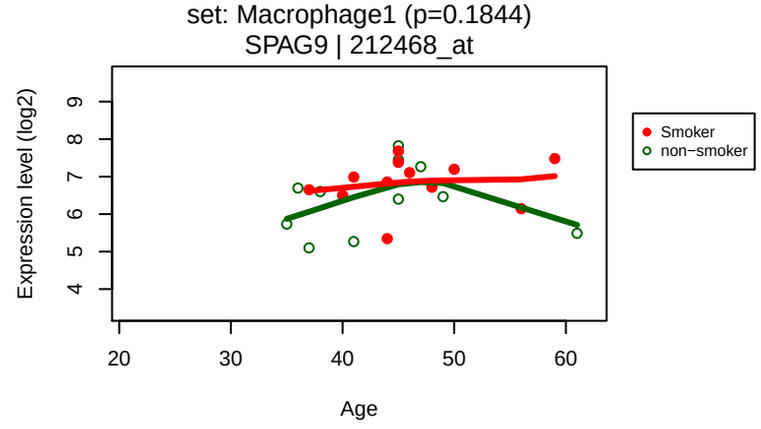

set: SmallAirway1 (p=0.6325)
SPAG9 | 212468_at
Expression level (log2)
Age
Smoker
non−smoker

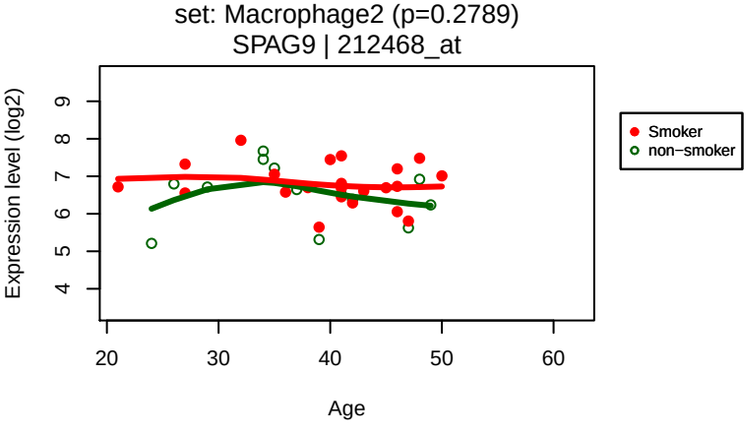

set: SmallAirway2 (p=0.025)
SPAG9 | 212468_at
Expression level (log2)
Age
Smoker
non−smoker

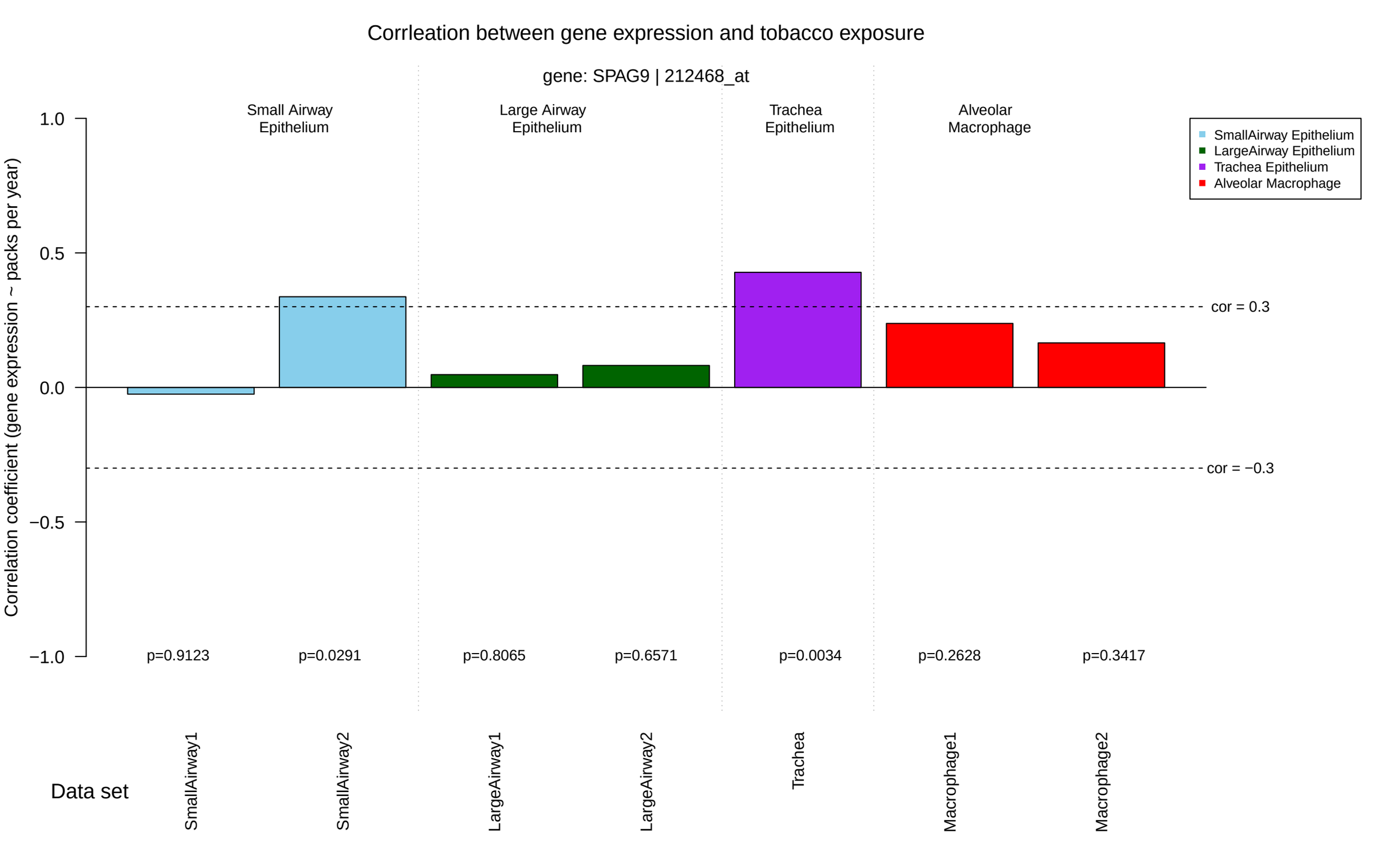

set: LargeAirway1 (p=0.5617)
SPAG9 | 212468_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5956)
SPAG9 | 212468_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0016)
SPAG9 | 212468_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1844)
SPAG9 | 212468_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2789)
SPAG9 | 212468_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPAG9 | 212468_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9123
p=0.0291
p=0.8065
p=0.6571
p=0.0034
p=0.2628
p=0.3417
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2